

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

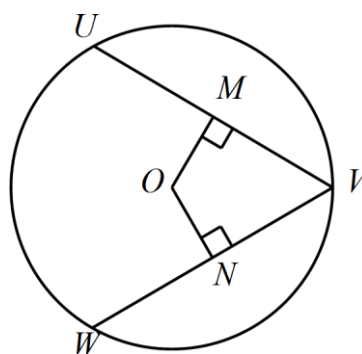
Use the multiple choice answer sheet for Questions 1-10.

1 The expansion of $(2x - y)^3$ is

- (A) $8x^3 - 4x^2y + 4xy^2 - y^3$
- (B) $8x^3 - 12x^2y + 6xy^2 - y^3$
- (C) $8x^3 - 4x^2y - 2xy^2 - y^3$
- (D) $8x^3 - 12x^2y - 6xy^2 + y^3$

2 O is the centre of the circle. If $OM = ON$, which of the following is true?

- (I) $UV = VW$
 - (II) $MV = NV$
- (A) (I) only
 - (B) (II) only
 - (C) Both (I) and (II).
 - (D) Neither (I) nor (II).



3 The exact value of $\tan\left(\sin^{-1}\left(\frac{-2}{3}\right)\right)$ is:

- (A) $\frac{-2}{\sqrt{5}}$
- (B) $\frac{2}{\sqrt{5}}$
- (C) $\frac{-2}{3}$
- (D) $\frac{2}{5}$

4 $\int \frac{2}{\sqrt{4-25x^2}} dx$ is:

- (A) $\frac{2}{5} \sin^{-1} \frac{x}{2} + C$
- (B) $\frac{2}{5} \sin^{-1} \frac{5x}{2} + C$
- (C) $\frac{2}{5} \sin^{-1} \frac{x}{10} + C$
- (D) $2 \sin^{-1} \frac{2x}{5} + C$

- 5 A particle moves in a straight line such that its velocity, v is given by $v = \cos 2x$ when at displacement x from the origin O .

The acceleration of the particle is given by

- (A) $-2 \sin 2x$
- (B) $-\sin 2x \cos 2x$
- (C) $\frac{1}{2} \sin 2x$
- (D) $-\sin 4x$

- 6 The horizontal asymptote of the function $y = \frac{2x^2 - 4x + 3}{3x^2 - 5}$ is

- (A) $y = -\frac{3}{5}$
- (B) $y = \frac{2}{3}$
- (C) $y = 0$
- (D) $y = 2$

- 7 The number of solutions of the equation $\cos 3x = -0.5$ for $0 \leq x \leq 2\pi$ is
- (A) 1
(B) 2
(C) 3
(D) 6
- 8 At a certain instant, a sphere is of radius 10 cm and the radius is increasing at a rate of 2 cm/s. The rate of increase of the volume of the sphere, in cm^3/s is
- (A) 80π
(B) $\frac{800\pi}{3}$
(C) 800π
(D) 400π
- 9 The point P divides the interval from $A(-1, 4)$ to $B(3, 7)$ externally in the ratio 5:2. What is the y co-ordinate of P ?
- (A) 2
(B) 6
(C) $6\frac{1}{7}$
(D) 9
- 10 When the polynomial $x^3 + 2x^2 - 3x + k$ is divided by $(x + 2)$, the remainder is 3. The value of k is:
- (A) -3
(B) -7
(C) 9
(D) 13

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra booklets are available.
In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11. (15 marks). Use a Separate Booklet.	Marks
(a) Factorise $8x^3 - y^3$.	1
(b) Find the constant term in the expansion of $\left(x^2 + \frac{3}{x}\right)^6$.	2
(c) Given $f(x) = \frac{1}{7-x}$,	
(i) Find $f^{-1}(x)$.	1
(ii) State the domain of $f^{-1}(x)$.	1
(iii) State the range of $f^{-1}(x)$.	1
(d) Solve $\frac{x^2 - 4}{x} > 0$	3

Question 11 continued on next page.

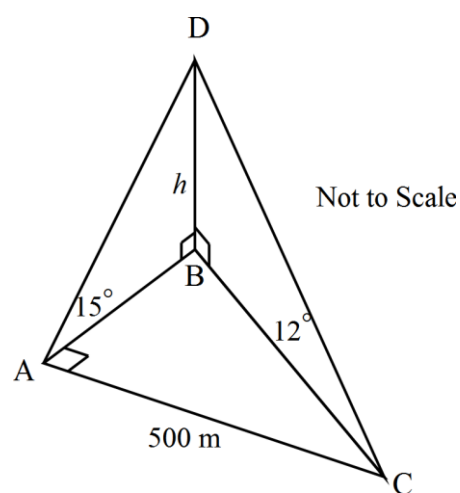
- (e) Evaluate $\int_1^{\sqrt{2}} x\sqrt{x^2-1} \, dx$ using the substitution $u = x^2 - 1$.

3

- (f) From a point A, the top of a tower BD, which is directly north of A, has an angle of elevation of 15° . After walking 500 metres on a bearing of 090° , the top of the tower has an angle of elevation of 12° .

3

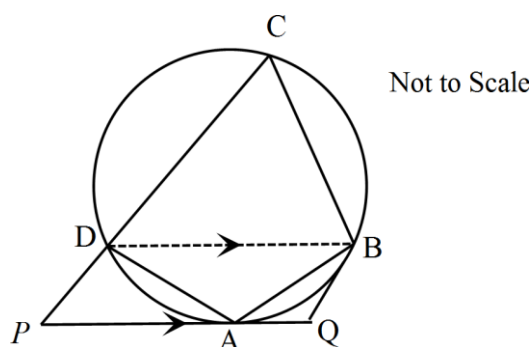
Find the height of the tower, to the nearest metre.



(a) Prove $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$.

3

(b)



Let $ABCD$ be a cyclic quadrilateral. The tangents from Q touch the circle at A and B . The diagonal DB is a parallel to the tangent AQ and QA produced intersects with CD produced at P . Copy this diagram into your booklet.

Let $\angle QAB = \alpha$.

(i) Prove that $\triangle BAD$ is isosceles.

2

(ii) Explain why $\angle BCD = 2\alpha$.

2

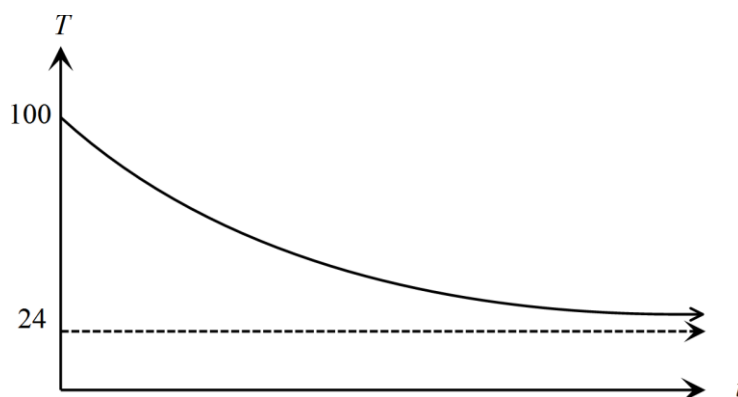
(iii) Show that P, Q, B and C are concyclic points.

2

Question 12 continued on next page.

- (c) The graph below represents the relationship between T , the temperature in $^{\circ}\text{C}$ of a cooling cup of tea, and t , the time in minutes.

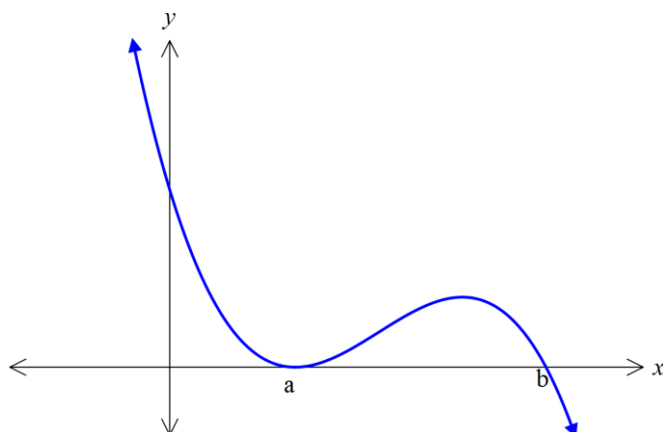
The rate of cooling of this tea is given by $\frac{dT}{dt} = -k(T - A)$ where k and A are constants and $k > 0$.



- | | | |
|-------|--|---|
| (i) | Show that $T = A + Be^{-kt}$ is a solution to the differential equation where B is a constant. | 1 |
| (ii) | By considering the graph at $t = 0$ and when $t \rightarrow \infty$, find the values of A and B . | 2 |
| (iii) | If the temperature of the tea is 40°C after 75 minutes, show that $k = -\frac{1}{75} \ln\left(\frac{4}{19}\right)$. | 2 |
| (iv) | Hence, find the rate at which the tea is cooling after 75 minutes. | 1 |

- (a) The graph of a polynomial, $P(x)$ is shown below.
Write down a possible equation for $P(x)$.

1



- (b) The polynomial equation $x^3 - 11x^2 + cx - 36 = 0$ has 3 positive roots. One root is the product of the other two roots.

(i) Show that 6 is a root of the polynomial

2

(ii) Hence, find the value of c .

2

- (c) Consider the function $f(x) = x^2 + 3\ln(x-2)$

(i) By sketching $y = -x^2$ and $y = 3\ln(x-2)$ on the same axes, show that $f(x) = 0$ has exactly one solution, α , where $2 < \alpha < 3$.

3

(ii) Taking $x = 2.2$ as the first approximation, use one application of Newton's method to find a second approximation to α . Give your answer to 3 decimal places.

2

- (d) (i) Show that the equation of the normal to the parabola $x^2 = 16y$ at the point $P(8p, 4p^2)$ is given by $x + py = 8p + 4p^3$.

2

(ii) Find, in terms of p , the distance SN , where S is the focus of the parabola and N is the point where the normal intersects the y -axis.

2

(iii) Hence find the area of $\triangle PSN$ in terms of p .

1

(a) $f(x) = \cos^{-1}(\cos x), -\pi \leq x \leq \pi.$

(i) Differentiate $f(x)$ and hence find the possible values of $f'(x)$. **3**

(ii) Hence or otherwise, sketch $f(x)$. **1**

(b) The temperature $T^{\circ}\text{C}$ of an unheated building is modelled using the equation
$$T = 23 + 2\cos\left(\frac{\pi t}{12}\right) + 5\sin\left(\frac{\pi t}{12}\right), \quad 0 \leq t \leq 24$$
 where t is the number of hours after midday.

(i) Express $2\cos\theta + 5\sin\theta$ in the form $R\cos(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. **2**
Give the values of R and α to 3 significant figures.

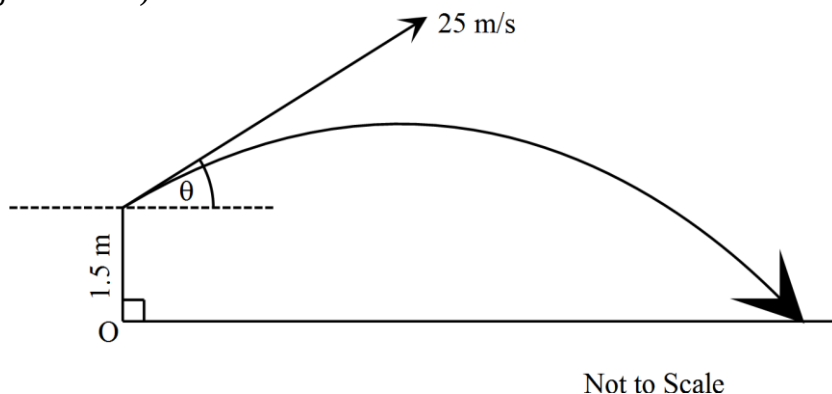
(ii) Hence, calculate the minimum temperature of the building and the time when this minimum occurs. **2**

Question 14 continued on next page.

- (c) An athlete releases a javelin with speed 25 m/s at an angle θ to the horizontal from a height 1.5 m above the ground.

Take the origin to be the point on the ground directly below the point of release, as shown. You may assume the horizontal position of the javelin, at time t , is given by $x = 25t \cos \theta$. DO NOT PROVE THIS.

(take $g = 10 \text{ ms}^{-2}$).



- (i) Show that the height, y of the javelin at time t is given by 2
 $y = -5t^2 + 25t \sin \theta + 1.5$.
- (ii) The school record for the javelin throw is 60 m. Show that the athlete achieves a school record throw when 3
 $288 \tan^2 \theta - 600 \tan \theta + 273 = 0$.
- (iii) Hence, calculate in what range θ must lie for the athlete to exceed the school record. 2

End of Paper

SOLUTIONS

1. B
2. C
3. A
4. B
5. D
6. B
7. D
8. C
9. D
10. A

d) $x^2 \times \frac{x^2-4}{x} > 0 \times x^2$
 $x(x-2)(x+2) > 0$

$\therefore -2 < x < 0, x > 2$

①

①

①

QUESTION 11

(a) $8x^3 - y^3 = (2x)^3 - y^3$
 $= (2x-y)(4x^2 + 2xy + y^2)$

Im RLW.

(b) $(x^2 + \frac{3}{x})^6$

General term = ${}^6C_k (x^2)^{6-k} (\frac{3}{x})^k$

$= {}^6C_k x^{12-2k} \frac{3^k}{x^k} x^{-k}$
 $= {}^6C_k x^{12-3k} 3^k$

For constant term, $12-3k=0$

$k=4$ - ①

$\therefore$ constant term = ${}^6C_4 \cdot 3^4$
 $= 1215$ - ①

QUESTION 11 (CONT)

(c) (i) For inverse $x = \frac{1}{7-y}$

$7x - xy = 1$

$xy = 7x - 1$

$y = \frac{7x-1}{x}$

$\therefore f^{-1}(x) = \frac{7x-1}{x}$

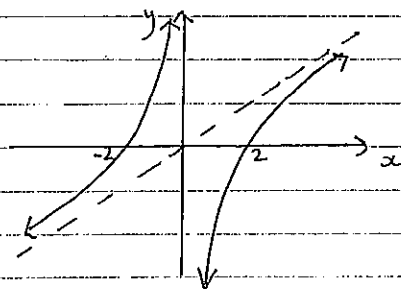
1 RLW.

(ii) domain $f^{-1}(x)$: all real $x \neq 0$ Im RLW

(iii) range $f^{-1}(x)$: all real $y \neq 7$ Im RLW.

(d) $\frac{x^2-4}{x} > 0$

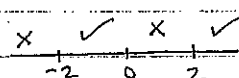
Consider $y = \frac{x^2-4}{x}$
 $= x - \frac{4}{x}$



From graph $\frac{x^2-4}{x} > 0$ when $x > 2$ or $-2 < x < 0$

or critical points when $x=0$

or $\frac{x^2-4}{x} = 0$



$x = \pm 2$

Test $x = -3$ LHS = $\frac{5}{-3} < 0$ FALSE $\therefore$ solution
 $x = -1$ LHS = $3 > 0$ TRUE $-2 < x < 0$ or
 $x = 1$ LHS = $-3 < 0$ FALSE $x > 2$

- ① - critical values
- ① - test / graph
- ① - solution.

QUESTION 12 (CONT.)

(b) (i) $\begin{cases} \therefore AD = AB & (\text{equal sides opposite equal angles}) \\ \therefore \Delta ABD \text{ isosceles} & (\text{in } \Delta ABD) \end{cases}$

$$\angle DAB = 180 - \angle ADB - \angle ABD$$

(ii) $\angle DAB = 180 - \alpha - \alpha$ (angle sum of $\Delta BAD = 180^\circ$)
 $= 180 - 2\alpha$ — ①

$\angle BCD = 180 - \angle DAB$ (opposite angles of cyclic quadrilateral $ABCD$ are supplementary)
 $= 180 - (180 - 2\alpha)$ ①
 $= 2\alpha$ as required.

(iii) $\angle BCD = \angle BCP$ (common)
 $= 2\alpha$ (from ii)

$AO = OB$ (tangents from external point S to circle at A and B are equal)

$\therefore \Delta BAO$ is isosceles.

$\angle OAB = \angle OBA = \alpha$ (angles opposite equal sides in ΔBAO)

$\therefore \angle AOB = 180 - 2\alpha$ (angle sum of $\Delta AOB = 180^\circ$) — ①
 $= \angle POB$ (common)

Now $\angle POB + \angle PCB$

$= 180 - 2\alpha + 2\alpha$ — ①

$= 180$

$\therefore \angle POB$ and $\angle PCB$ are supplementary

$\therefore POBC$ are concyclic (opposite angles in cyclic quadrilateral are supplementary)

QUESTION 12 (CONT.)

(c) (i) $T = A + Be^{-kt}$ — ①
 $\frac{dT}{dt} = -kBe^{-kt}$

From ①, $Be^{-kt} = T - A$ ← must show this or equivalent.

$\therefore \frac{dT}{dt} = -k(T - A)$

$\therefore$ it is a solution of differential equation.

(ii) When $t=0$, $T = A + B$ (from ①)

From graph $T = 100$

$\therefore A + B = 100$

As $t \rightarrow \infty$ $T \rightarrow A$ (from ①)

From graph, $T \rightarrow 24$

$\therefore A = 24$ — ①

$B = 76$ — ①

(iii) $T = 24 + 76e^{-kt}$

When $t = 75$ $T = 40$

$40 = 24 + 76e^{-75k}$ — ①

$16 = 76e^{-75k}$

76

$-75k = \ln\left(\frac{16}{76}\right)$

$k = -\frac{1}{75} \ln\left(\frac{4}{19}\right)$

QUESTION 12 (CONT)

$$(iv) \frac{dT}{dt} = -K(T - 24)$$

$$= -K \times (40 - 24)$$

$$= \frac{1}{75} \ln \left(\frac{4}{19} \right) \times 16$$

$$= \frac{16}{75} \ln \left(\frac{4}{19} \right) \text{ degrees/minute}$$

$$= -0.332 \text{ (3sf) degrees/min.}$$

$$\therefore \text{rate of cooling } 0.332 \text{ deg/min.} \quad - \textcircled{1}$$

QUESTION 13

$$(a) P(x) = -K(x-a)^2(x-b) \quad (\text{any equation of this form})$$

In RLW.

$$\text{or } K(x-a)^2(b-x)$$

$$(b) x^3 - 11x^2 + cx - 36 = 0$$

$$\text{Roots } \alpha, \beta, \alpha\beta \quad \alpha, \beta > 0$$

$$\begin{array}{llll} \text{(i) sum of roots} & \alpha + \beta + \alpha\beta = 11 & (-b/a) & \textcircled{1} \\ \text{sum (in pairs)} & \alpha\beta + \alpha^2\beta + \alpha\beta^2 = c & (c/a) & \textcircled{2} \\ \text{sum (in threes)} & \alpha^2\beta^2 = 36 & (-d/a) & \textcircled{3} \end{array} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \textcircled{4}$$

$$\therefore \alpha^2\beta^2 = 36 \quad - \textcircled{1}$$

$$\alpha\beta = 6 \quad \text{since } \alpha\beta > 0 \quad - \textcircled{1}$$

$$\therefore 6 \text{ is a root of polynomial.}$$

$$\text{(ii) Subst into } \textcircled{1} \quad \alpha + \beta + 6 = 11$$

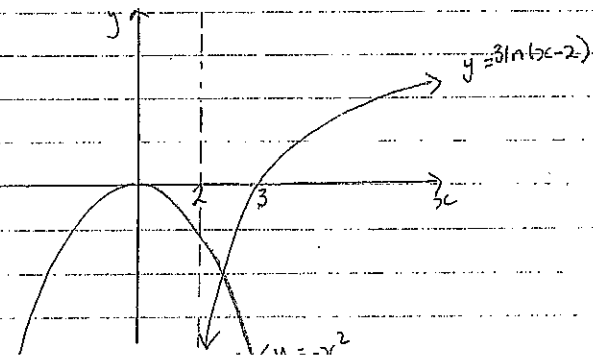
$$\alpha + \beta = 5 \quad - \textcircled{1}$$

$$\text{Subst into } \textcircled{2}, \quad \alpha\beta(1 + \alpha + \beta) = c$$

$$6(1 + 5) = c$$

$$c = 36. \quad - \textcircled{1}$$

(c)



From graph $y = -x^2$ and $y = 3\ln(x-2)$ have one point of intersection.

$\therefore 3\ln(x-2) = -x^2$ has one root
i.e. $3\ln(x-2) + x^2 = f(x)$ has one zero
i.e. $f(x) = 0$ has one solution.

From graph solution lies between 2 and 3

$\therefore$ If $f(x) = 0$ $2 < x < 3$.

$$(ii) \quad f(2.2) = 2.2^2 + 3\ln(2.2-2) \\ = 4.84 + 3\ln(0.2)$$

$$f'(x) = 2x + \frac{3}{x-2} \quad (1)$$

$$f'(2.2) = 4.4 + \frac{3}{0.2}$$

$$= 19.4.$$

Using Newton's method, second approximation $= x_0 - \frac{f(x_0)}{f'(x_0)}$

$$= 2.2 - \frac{4.84 + 3\ln(0.2)}{19.4}$$

$$= 2.19939 \dots \quad (1)$$

$$= 2.199 \quad (3dp)$$

QUESTION 13 (CONT)

$$(d) (i) \quad x^2 = 16y \quad \text{at} \quad P(8p, 4p^2)$$

$$y = \frac{x^2}{16}$$

$$\frac{dy}{dx} = \frac{x}{8}$$

$$\text{At } P, \quad \frac{dy}{dx} = \frac{8p}{8} = p$$

$$\therefore \text{gradient of normal} = -\frac{1}{p} \quad - (1)$$

$\therefore$ Eq of normal is

$$y - 4p^2 = -\frac{1}{p}(x - 8p) \quad - (2)$$

$$py - 4p^3 = -x + 8p$$

$$x + py = 8p + 4p^3 \quad (\text{as required}) \quad (*)$$

$$(ii) \quad 4a = 16$$

$$a = 4$$

$$\therefore \text{Focus } S = (0, 4)$$

N is where $x = 0$

$$\text{Sub in } (*), \quad py = 8p + 4p^3$$

$$y = 8 + 4p^2 \quad - (1)$$

$$N = (0, 8 + 4p^2)$$

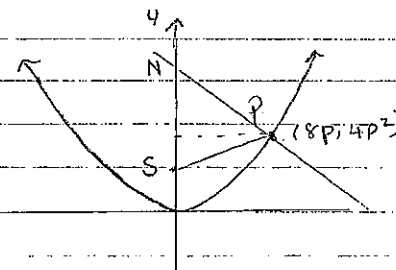
$$\therefore SN = 8 + 4p^2 - 4$$

$$= 4 + 4p^2 \quad - (1)$$

$$(iii) \quad \text{area } \triangle PSN = \frac{1}{2} \times b \times h$$

$$= \frac{1}{2} \times SN \times 8p$$

$$= 4p(4 + 4p^2) = 16p(1 + p^2) \quad (1) \text{ R/W}$$



QUESTION 14

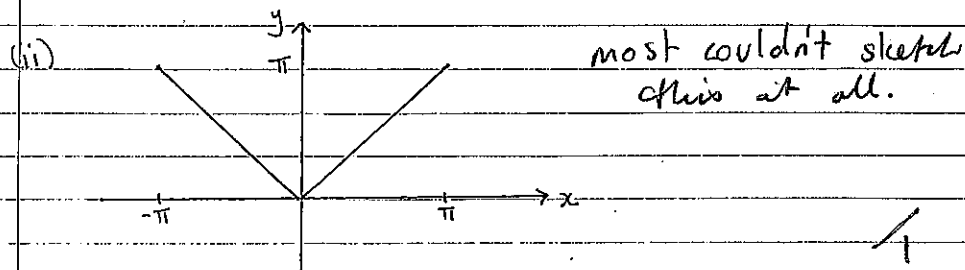
(a) $f(x) = \cos^{-1}(\cos x) \quad -\pi < x \leq \pi$

(i) $f'(x) = \frac{-1}{\sqrt{1-\cos^2 x}} \times -\sin x \quad \textcircled{1}$

$f'(x) = \frac{\sin x}{|\sin x|}$ ← few remembered $|\sin x|$

$\therefore f'(x) = \begin{cases} 1 & \text{if } \sin x > 0 \Rightarrow 0 < x < \pi \\ -1 & \text{if } \sin x < 0 \Rightarrow -\pi < x < 0 \end{cases} \quad \textcircled{1}$

$f'(x)$ not defined w.r. $x = 0, \pm\pi$.
 not done by many at all. /3



(b) (i) $2\cos\theta + 5\sin\theta = R\cos(\theta - \alpha)$
 $= R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$

$\therefore R\cos\alpha = 2 \quad R = \sqrt{2^2 + 5^2}$
 $R\sin\alpha = 5 \quad = \sqrt{29}$
 $= 5.39 \quad (3.s.f.)$

$\therefore \tan\alpha = \frac{5}{2}$
 $\alpha = 1.19 \quad (3.s.f.)$

0.379π

← checked here. /2

QUESTION 14 (CONT)

(b) (i) $\therefore 2\cos\theta + 5\sin\theta = 5.39\cos(\theta - 1.19)$

(ii) Minimum temp

$T = 23 + 2\cos\frac{\pi t}{12} + 5\sin\frac{\pi t}{12}$

$= 23 + 5.39\cos\left(\frac{\pi t}{12} - 1.19\right)$

Minimum occurs when $\cos\left(\frac{\pi t}{12} - 1.19\right) = -1$

$\therefore \text{minimum temp} = 23 - 5.39$
 $= 17.61^\circ \quad \textcircled{1}$

$\cos\left(\frac{\pi t}{12} - 1.19\right) = -1$

$\frac{\pi t}{12} - 1.19 = -\pi, \pi, 3\pi, \dots$

Now $0 \leq t \leq 24$

$-1.19 \leq \frac{\pi t}{12} - 1.19 \leq 5.093$

$\therefore \frac{\pi t}{12} - 1.19 = \pi$ /2

$t = 16.545 \dots \text{ hours}$

$= 16 \text{ hours } 33 \text{ minutes (nearest min)}$

$\therefore \text{time minimum occurs} = 04:33 \text{ am} \quad \textcircled{1}$

↑ needed

QUESTION 14 (CONT)

(c) (i) $\ddot{y} = -10$

$\dot{y} = -10t + C$

When $t=0$ $\dot{y} = 25 \sin \theta$

$\therefore C = 25 \sin \theta$

$\dot{y} = -10t + 25 \sin \theta$ (1)

$y = -5t^2 + 25t \sin \theta + C_1$

When $t=0$ $y = 1.5$

$\therefore C_1 = 1.5$

$\therefore y = -5t^2 + 25t \sin \theta + 1.5$ (2) - (1) must have some sub

2

(ii) When $x=60$

$60 = 25t \cos \theta$

$t = \frac{60}{25 \cos \theta}$

$t = \frac{12}{5 \cos \theta} = \frac{12 \sec \theta}{5}$ (1)

Subst $t = \frac{12 \sec \theta}{5}$ $y=0$ into (2)

$0 = -5 \left(\frac{12 \sec \theta}{5} \right)^2 + 25 \left(\frac{12 \sec \theta}{5} \right) \sin \theta + 1.5$

$0 = \frac{-144 \sec^2 \theta}{5} + 60 \tan \theta + 1.5$ (1) (x5)

$0 = -144 \sec^2 \theta + 300 \tan \theta + 7.5$ (x2)

$0 = -288 \sec^2 \theta + 600 \tan \theta + 15$

$0 = -288 (\tan^2 \theta + 1) + 600 \tan \theta + 15$ (1)

$0 = -288 \tan^2 \theta + 600 \tan \theta - 273$

$\therefore 288 \tan^2 \theta - 600 \tan \theta + 273 = 0$

3

need SEC to get to 2nd place

QUESTION 14 (CONT)

(c) (iii) $288 \tan^2 \theta - 600 \tan \theta + 273 = 0$

$\tan \theta = \frac{600 \pm \sqrt{600^2 - 4 \times 288 \times 273}}{2 \times 288}$

$= \frac{600 \pm 24\sqrt{79}}{576}$

$= \frac{25 \pm \sqrt{79}}{24}$

$\theta = 54.693... \text{ or } 33.8744... \text{ (} 0 < \theta < 90^\circ \text{)}$

$\therefore$ For athlete to exceed school record

$\boxed{33^\circ 53'} < \theta < \boxed{54^\circ 41'}$

- (1) correct inequality

my notes

2

54.69346

33.87445

1.412008101

0.6713252326

54° 41' 36.4"

33° 52' 28.05"

54° 41' ↓

↑ 33° 53'

ratians

0.95458069 ↓

0.59122083 ↑